

Skills Summary

Finding the Missing Factor

Use this reference while completing your worksheet or when studying.

What a missing factor is. In $\square \times b = c$ the box holds the *missing factor*: how many are in each equal group, or how many equal groups there are.

The rule: a missing factor is found by dividing. $\text{Total} \div \text{known factor} = \text{missing factor}$.

The check: multiply back. The two factors must make the total again.

1. Missing factor from an equal-groups picture

Count one group, not the whole picture. Then ask which number the picture is missing: the number of groups, or the number in each group. Write the multiplication sentence first, and let the division fact fill the box.

Example: 5 equal rows of counters, 30 counters in all. How many counters are in each row?

Step 1. The rows are equal, so this is groups times size, and the size of a row is what is missing.

Step 2. Divide the total by the number of groups: $30 \div 5$, and check it with $5 \times 6 = 30$.

Each row holds **6** counters.

Try it: 3 equal baskets hold 21 apples in all. How many apples are in one basket?

check: 7

2. Missing factor in a number sentence

With no picture, use the fact family. Every multiplication sentence carries two division facts, and one of them names the box directly.

A division sentence with a missing divisor is the same question in disguise: rewrite it as a multiplication sentence before you divide.

Example: Find the missing factor in $6 \times \square = 48$.

Step 1. One factor and the total are known, so the matching division fact names the missing factor.

Step 2. Divide the total by the known factor: $48 \div 6$, then check with $6 \times 8 = 48$.

The missing factor is **8**.

Try it: Find the missing number in $40 \div \square = 8$.

check: 5

3. Missing factor in a story

Read for the two group words first: how many groups, and how many in each group. One of them is missing — that is your box.

Write the equation before computing, and finish the answer with the unit the story uses, such as *rows*, *packs*, or *chairs*.

Example: A shop puts 54 apples into 6 equal crates. How many apples are in each crate?

Step 1. The crates are equal groups and the number in one crate is missing, so the story is groups times size equals total.

Step 2. The equation is $6 \times \square = 54$, so divide: $54 \div 6$, and check with $6 \times 9 = 54$.

Each crate holds **9** apples.

Try it: A baker fills trays with 7 rolls each and uses 35 rolls in all. How many trays does she fill?

check: 5

(!) Watch out: subtracting the two numbers in the sentence is the most common mistake, and it never answers a missing-factor question. Always finish by multiplying your answer back to the total.